Tribhuvan University

Faculty of Humanities & Social Sciences

OFFICE OF THE DEAN

Bachelor in Computer Applications

Course Title: **Information Security**

Code No: Semester: **5**

Full Marks: **60**   Pass Marks: **24**   Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: *Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.*

Group B

1. What is hash function? Describe how 128-bit of hash value is generated by taking an input message of variable size using MD5 algorithm?

2. Explain different ethical issues in computing? Explain RSA algorithm with suitable numerical example.

3. What is symmetric cryptography? Explain round operation of DES algorithm. Describe Sub-Key generation process for DES rounds.

4. Define authentication. How Biometric information can be used for authentication? Explain.

5. What is access control? Explain attribute based access control with example.

6. Decrypt the cipher text "CRHG" using the Hill cipher with the key matrix3455

7. What is security risk assessment? What are different aspects of a successful security risk assessment?

8. Differentiate between virus, worm and Trojan horse.

9. What is Security threat and attack? Describe different types of attacks in brief.

10. Explain Diffie Hellman Key Exchange Protocol with suitable example.

11. Define Euler Toient function. Determine whether 37 is Composite or not using Miller Rabin Primality testing.

12. Write Short Notes on:a. Phishing Attackb. Two Factor Authentication